

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Sixth Semester B.E. (AIML) (Credit System) Degree Examinations**

May - June 2023

20AM604 – AUTONOMOUS SYSTEMS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	Define Autonomous Systems. Explain in detail examples for AS.	10	L*2	1	1
	b)	Define Robot? Mention the different types of robots?	05	L2	2	1
	c)	Differentiate between hard automation and soft automation.	05	L2	2	1
2.	a)	Explain the four software components for autonomous vehicle system architecture.	10	L2	1	1
	b)	Explain in detail overview of robots.	10	L2	2	1
3.	a)	List out and explain the sensors used in the Autonomous Systems?	10	L2	1	1
	b)	Differentiate between Robotics systems and Autonomous system.	06	L2	2	1
	c)	Mention in detail specifications of the robot.	04	L2	2	1
		Unit – II				
4.	a)	Explain with the diagram basic architecture of industrial robot.	10	L2	3	1
	b)	Write a note on : i)Ant colony algorithm ii)Firefly algorithm.	10	L2	4	1
5.	a)	Explain the different classes of robots according to the association JIRA and AFR.	06	L2	3	1
	b)	Write a note on robotic links with diagram.	04	L2	3	1
	c)	Explain A* algorithm with example.	10	L2	4	1
6.	a)	Mention the advantages of cartesian and cylindrical robots.	03	L2	3	1
	b)	Illustrate the joint symbols with the diagram.	04	L2	3	1
	c)	Define actuators? Represent its classification.	03	L2	3	1
	d)	Explain in detail improved Dijkstra's algorithm used in path planning.	10	L2	4	1
		Unit – III				
7.	a)	Explain with the block diagram components essential for the flight	10	L2	5	1
	b)	Differentiate between multi rotor and fixed wing.	06	L2	5	1
	c)	Explain with diagram degrees of freedom of a quadcopter.	04	L2	5	1
8.	a)	Write a note on the components of drone: i) Frame ii) PWM iii)PPM	12	L2	5	1
	b)	With the diagram explain the physical structure of drone in detail.	08	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
